

Student Report Generator using JavaScript

Overview

This project allows users to:

- Enter student details using a form
- Calculate total marks and percentage
- Generate a formatted student report
- Export the report as a PDF file

Project Structure

```
student-report-generator/  
|  
├─ index.html  
├─ style.css  
├─ script.js  
└─ README.md
```

index.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Student Report Generator</title>  
  
  <link rel="stylesheet" href="style.css">  
  
  <!-- jsPDF Library -->  
  <script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/  
jspdf.umd.min.js"></script>  
</head>  
<body>  
  
<div class="container">  
  <h1>Student Report Generator</h1>
```

```

<div class="form-section">
  <input type="text" id="studentName" placeholder="Student Name">
  <input type="text" id="rollNumber" placeholder="Roll Number">

  <input type="number" id="maths" placeholder="Maths Marks">
  <input type="number" id="science" placeholder="Science Marks">
  <input type="number" id="english" placeholder="English Marks">
  <input type="number" id="history" placeholder="History Marks">
  <input type="number" id="computer" placeholder="Computer Marks">

  <button onclick="generateReport()">Generate Report</button>
  <button onclick="downloadPDF()">Download PDF</button>
</div>

<div id="reportCard" class="report-card">
  <h2>Student Report</h2>

  <p><strong>Name:</strong> <span id="rName"></span></p>
  <p><strong>Roll Number:</strong> <span id="rRoll"></span></p>

  <table>
    <thead>
      <tr>
        <th>Subject</th>
        <th>Marks</th>
      </tr>
    </thead>
    <tbody id="marksTable"></tbody>
  </table>

  <p><strong>Total:</strong> <span id="totalMarks"></span></p>
  <p><strong>Percentage:</strong> <span id="percentage"></span>%</p>
  <p><strong>Grade:</strong> <span id="grade"></span></p>
</div>
</div>

<script src="script.js"></script>

</body>
</html>

```

style.css

```
body {
  font-family: Arial, sans-serif;
  background: #f4f6f8;
  margin: 0;
  padding: 20px;
}

.container {
  max-width: 800px;
  margin: auto;
  background: white;
  padding: 30px;
  border-radius: 10px;
  box-shadow: 0 0 10px rgba(0,0,0,0.1);
}

h1, h2 {
  text-align: center;
}

.form-section {
  display: grid;
  gap: 10px;
  margin-bottom: 30px;
}

input {
  padding: 10px;
  font-size: 16px;
}

button {
  padding: 12px;
  background: #007bff;
  color: white;
  border: none;
  cursor: pointer;
  border-radius: 5px;
}

button:hover {
  background: #0056b3;
}
```

```

.report-card {
  border-top: 2px solid #ddd;
  padding-top: 20px;
}

table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 15px;
}

table, th, td {
  border: 1px solid #ccc;
}

th, td {
  padding: 10px;
  text-align: center;
}

```

script.js

```

function generateReport() {
  const name = document.getElementById('studentName').value;
  const roll = document.getElementById('rollNumber').value;

  const subjects = {
    Maths: parseInt(document.getElementById('maths').value || 0),
    Science: parseInt(document.getElementById('science').value || 0),
    English: parseInt(document.getElementById('english').value || 0),
    History: parseInt(document.getElementById('history').value || 0),
    Computer: parseInt(document.getElementById('computer').value || 0)
  };

  let total = 0;

  const tableBody = document.getElementById('marksTable');
  tableBody.innerHTML = '';

  for (let subject in subjects) {
    total += subjects[subject];

    const row = `
      <tr>

```

```

        <td>${subject}</td>
        <td>${subjects[subject]}</td>
    </tr>
`;

    tableBody.innerHTML += row;
}

const percentage = (total / 500) * 100;

let grade = 'F';

if (percentage >= 90) {
    grade = 'A+';
} else if (percentage >= 80) {
    grade = 'A';
} else if (percentage >= 70) {
    grade = 'B';
} else if (percentage >= 60) {
    grade = 'C';
} else if (percentage >= 50) {
    grade = 'D';
}

document.getElementById('rName').innerText = name;
document.getElementById('rRoll').innerText = roll;
document.getElementById('totalMarks').innerText = total;
document.getElementById('percentage').innerText = percentage.toFixed(2);
document.getElementById('grade').innerText = grade;
}

function downloadPDF() {
    const { jsPDF } = window.jspdf;

    const doc = new jsPDF();

    const name = document.getElementById('rName').innerText;
    const roll = document.getElementById('rRoll').innerText;
    const total = document.getElementById('totalMarks').innerText;
    const percentage = document.getElementById('percentage').innerText;
    const grade = document.getElementById('grade').innerText;

    doc.setFontSize(18);
    doc.text('Student Report Card', 20, 20);

    doc.setFontSize(12);
    doc.text(`Student Name: ${name}`, 20, 40);
    doc.text(`Roll Number: ${roll}`, 20, 50);

```

```
doc.text(`Total Marks: ${total}`, 20, 60);
doc.text(`Percentage: ${percentage}%`, 20, 70);
doc.text(`Grade: ${grade}`, 20, 80);

doc.save(`${name}_Report.pdf`);
}
```

README.md

```
# Student Report Generator

## Features

- Enter student details
- Calculate percentage automatically
- Generate grade
- Download report as PDF
- Responsive UI

## Technologies Used

- HTML
- CSS
- JavaScript
- jsPDF

## How to Run

1. Download the project
2. Open `index.html` in browser
3. Enter student details
4. Click Generate Report
5. Click Download PDF
```

Future Enhancements

- Add database support
- Add multiple student report management
- Add charts and analytics
- Add teacher comments
- Add digital signature support

- Export reports to Excel